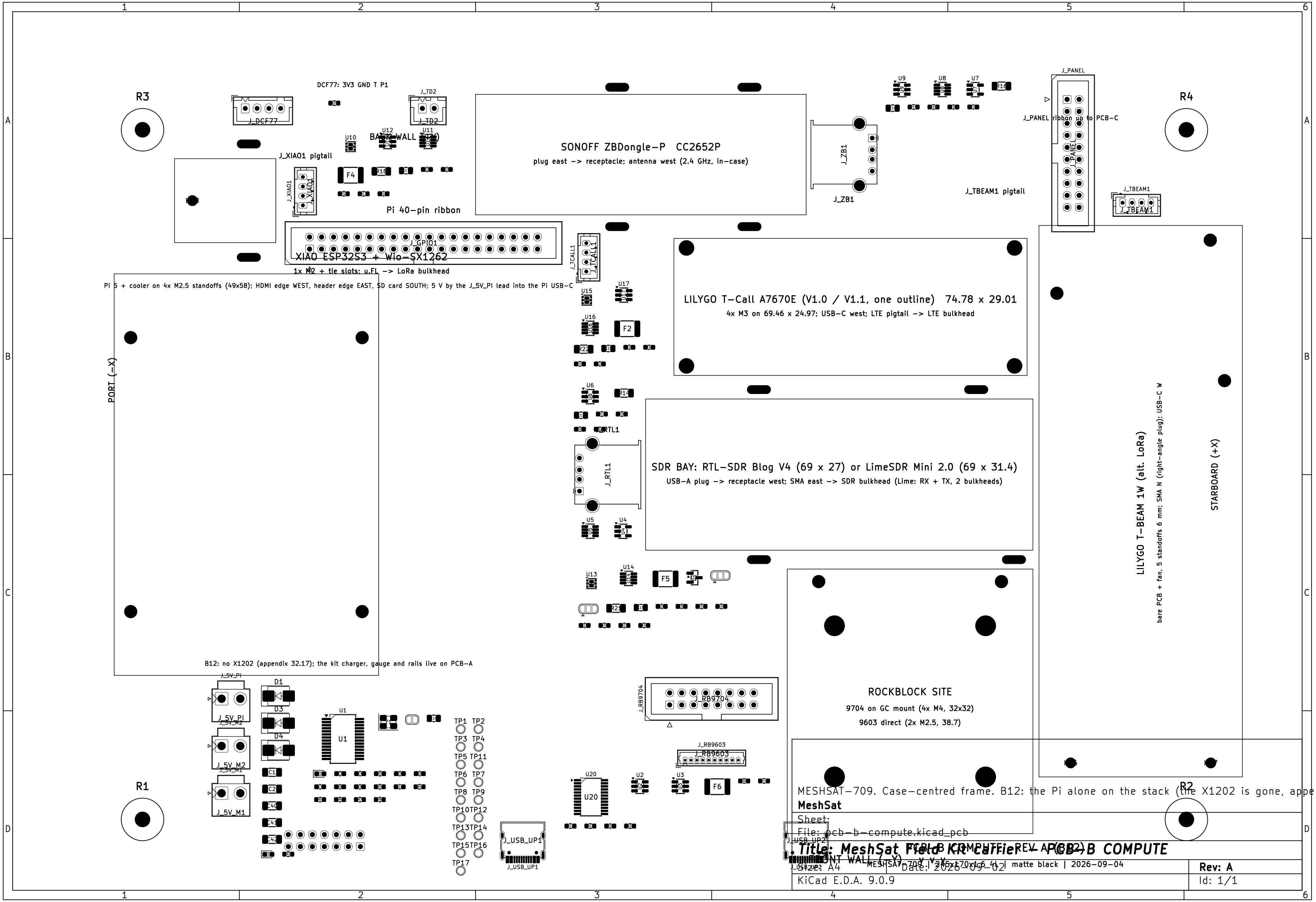
1
2
3
4
5
6
A
B
C
D
R3
R4
R1
R2
DCF77: 3V3 GND T P1
J_DCF77
J_TD2
U10
U12
U11
BACK WALL (+Y)
J_XIAO1 pigtail
J_XIAO1
F4
R18
Pi 40-pin ribbon
J_GPIO1
XIAO ESP32S3 + Wio-SX1262
1x M2 + tie slots; u.FL -> LoRa bulkhead
Pi 5 + cooler on 4x M2.5 standoffs (49x58); HDMI edge WEST, header edge EAST, SD card SOUTH; 5 V by the J_5V_PI lead into the Pi USB-C
PORT (-X)
B12: no X1202 (appendix 32.17); the kit charger, gauge and rails live on PCB-A
SONOFF ZBDongle-P CC2652P
plug east -> receptacle; antenna west (2.4 GHz, in-case)
J_ZB1
U9
U8
U7
R16
J_PANEL
J_PANEL ribbon up to PCB-C
J_TBEAM1 pigtail
J_TBEAM1
J_TCALL1
U15
U17
U16
F2
R22
LILYGO T-Call A7670E (V1.0 / V1.1, one outline) 74.78 x 29.01
4x M3 on 69.46 x 24.97; USB-C west; LTE pigtail -> LTE bulkhead
U6
R14
J_RTL1
SDR BAY: RTL-SDR Blog V4 (69 x 27) or LimeSDR Mini 2.0 (69 x 31.4)
USB-A plug -> receptacle west; SMA east -> SDR bulkhead (Lime: RX + TX, 2 bulkheads)
U5
U4
LILYGO T-BEAM 1W (alt. LoRa)
bare PCB + fan, 5 standoffs 6 mm; SMA N (right-angle plug); USB-C W
STARBOARD (+X)
U13
U14
F5
R21
J_RB9704
ROCKBLOCK SITE
9704 on GC mount (4x M4, 32x32)
9603 direct (2x M2.5, 38.7)
J_RB9603
J_5V_PI
J_5V_M2
J_5V_M1
D1
D3
D4
C1
C2
U1
TP1
TP2
TP3
TP4
TP5
TP11
TP6
TP7
TP8
TP9
TP10
TP12
TP13
TP14
TP15
TP16
TP17
J_USB_UP1
U20
U2
U3
F6
J_USB_UP2

MESHSAT-709. Case-centred frame. B12: the Pi alone on the stack (the X1202 is gone, appendix 32.

MeshSat

Sheet:

File: pcb-b-compute.kicad_pcb

Title: MeshSat Field Kit carrier — PCB-B COMPUTE

PCB-B COMPUTE: REV A (B12)

FRONT WALL (-Y)

MESHSAT-709 | 245x170x1.6 4L | matte black | 2026-09-04

Size: A4

Date: 2026-09-02

Rev: A

KiCad E.D.A. 9.0.9

Id: 1/1